Temporal Attention Neurons: Separating History Sensitivity, Addressing, and Recoverable Memory

Li Zexu

3 November 2026 — research manuscript draft

1 Abstract

A temporal attention mechanism can change a neuron's response to recent history without making that history recoverable through its permitted output. We synthesize the recorded Temporal Attention Neuron (TAN) research into an evidence-based account of this distinction. The evidence spans static addressing, native finite-window dynamics, task-specific response collisions, a controlled readout-training intervention, and comparisons with ordinary adaptive filters. Static scalar metric addressing correctly identifies all 6,705 intermediate-key queries in the archived noiseless three-key condition, whereas the signed scalar-product rule identifies none. This positive result presupposes an available key–value array. In a separate native temporal task, 545 target–distractor swap pairs with different labels have identical complete allowed responses, despite different ordered internal windows. A later clean-only training intervention improves mean clean balanced accuracy from 32.5881% to 38.1199%, but does not eliminate this structural limitation. In a matched-calibration detection experiment, TAN's balanced accuracy exceeds an ordinary adaptive comparator under independent noise but falls below it under stationary AR(0.9) noise. We distinguish these locally executed records from exploratory Chat-sandbox constructions that have not been independently replicated here. The resulting contribution is a task- and resource-specific account of what the architecture computes, where information becomes inaccessible, and which apparent gains depend on readout, training composition, or additional mechanisms. No universal superiority, biological implementation, or independent experimental replication is established.

2 Introduction

Attention is a computation over a representation, not by itself a guarantee that the representation retains the information required by a task. The Transformer makes content-dependent weighting operational at network scale [1]. TAN asks a smaller architectural question: what can a discrete dynamical unit do when its membrane update receives a surprise-gated, attention-weighted summary of recent inputs? Answering this question requires separating at least four properties: history sensitivity, address selection, preservation of a key–value association, and recovery through a specified observation.

The original project developed the TAN model, its experimental tasks, and an architectural progression from state and geometry to routing, binding, and composition. Existing manuscripts already discuss these distinctions. This paper is a retrospective synthesis of the subsequent evidence, not a duplicate presentation of those manuscripts or a new experimental campaign. In particular, later evidence qualifies a simple monotonic capability ladder: a representation may contain a target while its output cannot expose it; a scalar metric can address an existing array without solving dynamic storage; and a training change can improve observed accuracy without repairing an information-losing map.

We organize the evidence around three questions. Which positive results survive an explicit account of inputs, state, and readout? Which negative results are mathematical obstructions within a fixed contract, rather than broad claims about neurons? Which performance differences remain when ordinary baselines and training

composition are considered? The answer is a set of bounded capabilities, not a single success or failure label for all implementations called TAN.

3 Materials and evidence provenance

3.1 Source hierarchy and prior manuscripts

The original source baseline is commit `85213f0d417d54e4f48b301ed1c0e4200ef3c421`. The screened research publication is commit `222caa0201feff1e37a06049aa44c0e59e1cbaba`, containing 680 archived scientific files in four collections: audit v11 (440), native eight-stage studies (110), exploratory studies (100), and the local clean-training control (30). The exact screened ZIP has 189,164,458 bytes and SHA-256 `ca8a2ae51b4e8f18ab341fa483ca9970a7a3077c5d4e4013e5df13a9811b6167`. Hash agreement establishes file identity, not scientific validity.

An inventory made before drafting records 174 existing manuscript/report sources. The principal earlier manuscript families are `snapshot/manuscript`, `snapshot/paper`, and `snapshot/paper2`. Later correction fragments under the stage-localization audit are amendments, not an additional independent study. We retain these sources unchanged and use the latest completed report relevant to each claim. For static addressing, the authoritative experiment is P1A Rev2, not the superseded first run. For clean training, the frozen completed protocol supersedes an earlier proposal whose prospective wording remains historical.

During final publication checks, main advanced to commit `65b0d2e9310a5790e653b77114bf4b6c7aa93e9c`, adding a TAN-III manuscript focused on decoder floors, carrier rank, and a proposed dendritic interpretation. We inspected that concurrent manuscript and retain it unchanged. The present synthesis is a separate evidence-scoped account centered on observable recovery and the training-composition control. Shared numerical findings are not counted as additional experiments. The proposed biological interpretation is not treated here as an experimentally demonstrated implementation.

The evidence ledger accompanying this manuscript gives repository-relative links and hashes for the sources actually used. Reported numerical findings below are historical results checked against retained evidence. Writing and building this paper does not constitute a new run of the underlying experiments.

3.2 Execution and attribution

The model and original project are attributed to Li Zexu, consistent with the existing repository. Subsequent audit design, coding, mathematical analysis, computational checking, and this synthesis involved AI assistance. This statement does not assign every later derivation or implementation exclusively to the human author. AI systems are not listed as authors; human review remains necessary before journal submission.

We distinguish three evidence levels. First, exact arguments establish consequences of explicitly stated mathematical maps. Second, the native and local audit collections record executed computations, saved inputs, outputs, and source provenance. Some include separately implemented checkers or isolated replays; these are computational verification within an AI-assisted workflow, not independent human replication. Third, the `exploratory/chat-*` collections record Chat-sandbox conceptual experiments, including same-agent checks and reruns. We have inspected their documentation but have not independently reproduced those experiments in the current environment. The paper is an unreviewed research draft, not peer-reviewed evidence.

4 Models, contracts, and analysis

4.1 Native finite-window TAN

For the W5 native model used in the temporal task, let H_t be the ordered window containing the current input and four preceding inputs, and let h_{t-1} be the previous post-reset membrane state. The principal operations are

$$\mu_t = \frac{1}{5}\sum_{z \in H_t} z, \qquad S_t = \max(0, x_t - \mu_t), \qquad q_t = 2S_t,$$

$$C_t = \sum_i \alpha_{ti} H_{ti}, \qquad A_t = \tanh(S_t) C_t, \qquad u_t = 0.5 h_{t-1} + A_t,$$

$$y_t = \mathbf{1}\{u_t > 0.5\}, \qquad h_t = (1 - y_t) u_t.$$

The attention weights use exponentials of $q_t H_{ti}$, with a small additive denominator constant. Implementation details matter. The native v8 collision proof uses the historical unshifted exponential normalization with $\delta = 10^{-9}$. The clean-training control uses the source's stabilized softmax: subtract the maximum score before exponentiation and then add the same denominator constant. Because of that constant, these implementations are not exactly algebraically identical. We do not silently transfer an exact floating-point result between them [E2, E3].

The full finite-window system is Markovian in (h_t, H_t). Earlier natural-collision results show that membrane alone is insufficient; they do not establish non-Markovianity of the full state. The five window values plus membrane are a 48-byte logical payload under binary64 storage, excluding object overhead, temporaries, and the external decoder.

4.2 Higher-moment sensitivity is not positional addressing

For the ideal normalization without δ, define

$$Z(q) = \sum_i e^{qH_i}, \qquad C(q) = \frac{d}{dq} \log Z(q).$$

A local expansion at $q = 0$ gives

$$C(q) = \mu + q\kappa_2 + \frac{q^2}{2}\kappa_3 + \cdots,$$

where κ_j are cumulants of the empirical window distribution. Thus native attention generally responds to information beyond mean and variance. A clipped second-order approximation such as $\mu + q\,\mathrm{Var}(H)$ is an ablation, not an exact substitute. This is an analytical property of exponential weighting, not evidence of a novel statistical identity or of improved task accuracy.

The exact deployed normalization retains δ, so the ideal expansion must not be quoted as an exact formula for its stored outputs. Moreover, the context remains invariant to a permutation of a fixed window when the query is unchanged. Sensitivity to the amplitude distribution can coexist with loss of target position. The ordered window can retain position while a pooled output discards it. This is consistent with the broader distinction between set-invariant computation and ordered representations [2]. The bundled upstream control code contains a second-order context ablation, but its presence alone does not establish a validated higher-moment performance advantage [E3].

4.3 Static address selection

P1A Rev2 grants the models the full stored key, value, and validity arrays. For distinct keys and an exact query $q = k_t$, the metric score

$$s_i = -(q - k_i)^2$$

has a unique maximum at the requested key. By contrast, a nonzero scalar multiplier applied to ordered keys selects an extreme key; zero produces a tie rather than unique intermediate-key identification. These are elementary score properties. They do not prove a lower bound on neural state dimension.

The vector-QK implementation's cosine score has the same hard ordering as the metric on the observed domain, where $|0.4(q-k)| < \pi$. Its agreement with metric routing therefore has a direct explanation. Soft-output comparisons require accounting for score curvature and temperature: equal numeric temperature is not equal local discrimination. Values and the final addition/subtraction operation are explicitly available resources [E1].

4.4 Observation collision and training control

In native Probe1, the target occurs at step 5, distractors at steps 6–8, query at step 9, and the continuation is zero. The only decoder observation is

$$F = (u_9, u_{10}, u_{11}, u_{12}, u_{13}, y_9, y_{10}, y_{11}, y_{12}, y_{13}).$$

The v8 study compares histories (a, b, c, d, q) and (b, a, c, d, q) with $a, c \in [0.615, 0.635]$ and $b, d, q \in [1.365, 1.385]$. Their target classes differ outside the original overlapping low/high amplitude region. One anchor, all 32 box corners, and 512 frozen random draws give 545 paired cases [E2].

The later training control asks a different question: whether the composition of readout training data contributes to weak clean-task accuracy. It holds the stabilized W5 source dynamics and 33-parameter linear readout fixed. For each of three historical seeds, clean-only and mixed arms each use 1,500 examples, balanced at 500 per class. The mixed arm contains 250 clean and 250 distractor examples per class. Its 750 clean examples are a subset of the clean-only arm. Selection uses a fixed permutation without consulting test values. Both arms evaluate the same 1,001 clean examples per seed. Each arm uses its own train-only standardization, zero initialization, L2 coefficient 10^{-4}, and the frozen L-BFGS-B procedure, without validation-based tuning [E3].

Balanced accuracy is the arithmetic mean of class recalls. Training uncertainty is assessed only conditionally: 2,000 paired bootstrap draws resample original test reuse-groups within each fixed seed, preserve pairing between arms, and average equally across seeds. This does not integrate over training-set, optimizer, or seed-selection uncertainty.

4.5 Matched-calibration adaptive baseline

The separate v11 detection task uses W10 TAN with signed surprise, a magnitude gate, leak 0.8, strict threshold 1, and zero initial state. It must not be confused with the W5 rectified-surprise classifier. The ordinary comparator has two persistent scalars:

$$m_t = 0.8m_{t-1} + 0.2x_t, \quad I_t = \kappa \left| \frac{9}{8}(x_t - m_t) \right|, \quad u_t = 0.8u_{t-1}^{\text{post}} + I_t.$$

Both models use the same threshold/reset and observation window. A separate 512-history independent-noise calibration fixes $\kappa = 0.840938900477691$ by matching pooled absolute effective current, not accuracy or test outcomes. The comparator is an engineering implementation of input adaptation, not a claimed reproduction of a particular biological neuron. Adaptation-based separation of fast signals and slower backgrounds already has primary-literature precedent [3].

The test contains 1,024 innovation histories paired across independent and stationary AR(0.9) noise, step/catch conditions, and both models. Each waveform lasts 60 updates, with step onset at 15. A detection is any spike during updates 15–19; the readout therefore knows onset. Misses receive capped delay 5. The bootstrap resamples whole innovation groups, preserving every pairing. Calibration is frozen. Matching one pooled current statistic does not match all state, computation, current variance, or physical energy [E4].

5 Results

5.1 Static addressing succeeds within its granted memory interface

In the archived noiseless three-key condition, scalar metric attention and vector QK each identify all 6,705 intermediate-key role queries, whereas the signed scalar rule identifies zero. In the two noisy conditions, both-address success is 9,876/10,000 and 7,986/10,000 for each addressing method. These are observed task outcomes, not unconditional robustness guarantees.

The retained verification recomputed 60,000 condition/trial records and 420,000 model-output rows from separately written score formulas. All hard choices agreed exactly; the largest soft arithmetic discrepancy was 2.58×10^{-14}. The conditions do not supply 60,000 independent histories because two noisy conditions share base histories and scaled noise draws. This is saved-ledger verification, not regeneration of the historical random experiment [E1]. The result supports static metric addressing and challenges an overly narrow scalar-product obstruction. It does not establish dynamic writing or compact key–value storage.

5.2 Native response collisions separate stored from recoverable content

A rational interval certificate gives a step-8 gated current above 0.528613674287 throughout the specified box. Both native membranes therefore reset. At query, the two windows contain the same multiset, the same current input, and equal pooled context. The subsequent zero continuation closes the gate, yielding the same response

$$F = (u_9, u_9/2, u_9/4, u_9/8, u_9/16, 0, 0, 0, 0, 0).$$

At the anchor, $u_9 \approx 0.33538361861604077$. All 545 native TAN response pairs were also bit-identical in the recorded floating implementation. The ordered windows nevertheless differed when queried [E2].

Model or intervention	Identical complete responses / 545 pairs	Different spike responses / 545 pairs
Native LIF	545	0
Native recency delay line	0	0
Native uniform gated context	0	545
Native TAN	545	0
TAN with step-8 voltage reset suppressed	0	0

Suppressing only the step-8 voltage reset restores analog separation but not spike-only separation. It changes the dynamical rule and is a causal control, not another successful run of unchanged TAN. The delay-line and uniform-context results are valuable negative controls against a universal advantage claim; they are not isolated single-component interventions on TAN.

For an equal mixture of each opposite-label pair, any classifier of only the common response incurs error at least one half. This is a restricted indistinguishability statement. The corresponding conditional bound under the ideal continuous original present-distractor generator is only 3.2×10^{-9}. The construction therefore cannot explain the magnitude of average historical classification error. It proves a failure mode, not its prevalence.

5.3 Clean training improves the fixed readout, but only modestly

All six fitted optimizers converged. The recorded features exactly matched the archived stabilized-source response. Table 2 reports balanced accuracy on the shared clean test subsets [E3].

Historical seed	Clean-only training (%)	Mixed training (%)	Difference (percentage points)
20260904	38.5541	32.9776	5.5765
20260905	38.0594	32.0142	6.0451
20260906	37.7462	32.7725	4.9737
Equal-seed mean	38.1199	32.5881	5.5318

The conditional 95% percentile interval for the mean difference is [3.3391, 7.6080] percentage points. All classes are predicted in both arms, but changes are not uniform: for seed 20260905, low-class recall under clean-only training is 11.51%, while other recalls improve. An aggregate improvement must not be described as improvement for every class.

This intervention identifies training composition, including its induced normalization, as a contributor under the frozen pipeline. It does not isolate gradient conflict from preprocessing, identify a Bayes-optimal decoder, or establish architectural superiority. Mean performance remains modest, near the three-class balanced chance reference of one third. Historical reused tests are not a new blind benchmark.

The clean-map question must also remain separate from the proven distractor-swap collision. The retained v8 certificate contains nonzero distractors. We did not locate a separate clean-only collision certificate in the 680-file publication and do not relabel that theorem as one. A poor clean linear classifier does not prove noninjectivity of the clean input-to-response map; conversely, better clean training does not prove that map is injective. Any additional clean-map collision claim requires its exact source, witness, normalization, allowed query and observation contract before it can be included as a verified result.

5.4 Correlated noise reverses a detection advantage

The complete primary detection outcomes are [E4]:

Noise	Model	Misses / 1,024	Catch alarms / 1,024	Balanced accuracy (%)	Mean capped delay
Independent	TAN	0	1	99.9512	0.560547
Independent	Adaptive	0	137	93.3105	0.103516
AR(0.9)	TAN	62	6	96.6797	0.930664
AR(0.9)	Adaptive	7	40	97.7051	0.301758

The primary TAN-minus-adaptive AR balanced-accuracy difference is -1.0254 percentage points, with a predeclared 97.5% paired percentile interval [-2.0020, -0.0488]. The change in advantage from independent to AR noise is -7.6660 points, interval [-9.0820, -6.2988]. Both go against the prespecified superiority direction.

TAN still has fewer catch false alarms under AR noise, but misses more steps and responds more slowly. This is an operating tradeoff, not absence of useful behavior. The archived checker reports verification of 10,244 trajectories and all spike decisions; it is a separately implemented computational check, not external replication. TAN stores ten input coordinates plus membrane, whereas the adaptive comparator stores two scalars. The experiment therefore does not establish resource-normalized superiority for TAN.

5.5 Modified carriers and exploratory constructions

The native eight-stage series illustrates how a conditional rescue can require a changed contract. A longer gate time constant reduces sensor-only composition error from 0.270803 to 0.008093 on 4,096 held-out histories. However, the successful system also uses value-dependent gate drive, a nonzero query content probe, two analog samples, a known-equation nonlinear inverse, fixed known keys/timing, and targets restricted to $[-0.5, 0.5]^2$. It is neither original P1B success nor general variable-key retrieval [E5].

The interpolation control makes another useful separation. Two real payload registers can encode the line through two supplied key–value pairs, enabling ideal querying without retaining separate per-key records after the second write. Its recorded clean composition error is approximately 5.08×10^{-16} on 512 histories. This is classical interpolation with an explicit controller, separate ports, division, and atomic overwrite, not a new associative-memory identity or a demonstrated native neuron. Finite precision, leakage, and resource costs materially alter the result [E6].

Exploratory Chat-sandbox packages address related but different constructions. The bounded write/hold/query circuit adds state and control resources; a same-resource EMA performs at least as well in its stated small comparison. Moment-based latest-value schemes rely on address promises, conditioning, admissibility tests, or refusal. They can preserve a response at a noisy observed address while returning an incorrect value at the true key. These are informative conceptual controls, not replications of original TAN, and have not been independently rerun for this manuscript [E7].

6 Discussion

The evidence supports an operational distinction between access to information and the mere presence of information in state. Native attention can be sensitive to higher moments of the input distribution while remaining permutation-invariant at a fixed query. A common reset can remove order-dependent membrane information before that pooled query occurs. Conversely, static metric addressing succeeds when the candidate array and bound values are granted. There is no contradiction: the tasks expose different resources and ask different questions.

The training control adds a second distinction, between a map's information content and the performance of a particular fitted decoder. Improving the training distribution can improve a decoder on a fixed representation. Such improvement does not invert an exact collision and does not certify sufficient representation. Likewise, the existence of one collision family does not explain all empirical error. Reporting both findings prevents either a purely architectural or a purely optimization-based explanation from outrunning the evidence.

The adaptive comparison shows why a named baseline and a declared matching rule are necessary. A large independent-noise advantage can reverse when temporal correlation changes, even without retuning. Fewer false alarms alone do not establish better detection, just as higher late activity alone does not establish better information transfer. Current integrals and logical state counts are useful accounting quantities, but they are not measurements of energy, biological cost, or hardware throughput.

The project-specific contribution of this synthesis is the integration of these verified boundaries and interventions. Exponential pooling, nearest-neighbor addressing, permutation invariance, interpolation, and adaptation are established ideas. The evidence does not justify a claim of a breakthrough, universal task superiority, a minimal neural state theorem, or a new general theory of attention.

7 Limitations and reproducibility

This paper is retrospective. Protocols described as frozen were saved before their respective recorded experiments; that does not make the present cross-study synthesis prospectively registered. Different TAN variants, normalization rules, tasks, readouts, and resource budgets cannot be pooled into a common success percentage. The clean-only control has three fixed historical seeds and conditional test resampling, not a population-level training study. The collision proof applies to its supported box and continuation, not arbitrary noise or all future inputs. Static addressing assumes available stored candidates. The adaptive comparison covers one calibration and one correlation change.

The public archive is screened rather than a byte-identical image of every original machine file. It retains numeric data and records public text/path transformations. Historical hashes are distinct from current public-file hashes. Some reproduction guards require the pristine source or documented adaptation in a disposable copy. Download verification and source syntax checks are not experimental replication. No new model fit, simulation sweep, or scientific experiment was run to write this manuscript.

Readers can inspect the 680-file snapshot, per-claim ledger, archived protocols and saved outputs. The accompanying validation script checks source hashes, extracts selected headline numbers directly from saved CSV/JSON, and validates the original snapshot inventory without training. The build script produces the PDF and LaTeX from the canonical Markdown manuscript. Its output should be interpreted as publication consistency checking. Independent human review, independent execution in a specified environment, wider task testing, and physical implementation remain future work.

8 Conclusion

TAN provides a concrete test bed for separating temporal sensitivity, static addressing, binding, and recoverable output. The positive static-address evidence, native response collision, modest training-composition gain, and reversed correlated-noise comparison jointly support a resource- and task-specific account. They do not support treating attention-like weighting as sufficient for reliable memory. The useful next step is to test a sharply specified mechanism and output contract, while preserving both the positive results and the limitations established here.

9 References

1. Vaswani, A., et al. (2017). *Attention Is All You Need*. Advances in Neural Information Processing Systems 30. Primary paper.
2. Zaheer, M., et al. (2017). *Deep Sets*. Advances in Neural Information Processing Systems 30. Primary paper.
3. Benda, J., Longtin, A., and Maler, L. (2005). Spike-frequency adaptation separates transient communication signals from background oscillations. *Journal of Neuroscience*, 25(9), 2312–2321. DOI: 10.1523/JNEUROSCI.4795-04.2005. Primary full text.

10 Archived evidence references

The identifiers below refer to project artifacts rather than externally published, peer-reviewed papers. Exact paths, hashes, and scope are provided in `evidence_ledger.json` and `EVIDENCE.md` beside this manuscript.

- **E1.** P1A Rev2 condition summary; metric-address proposition; v8 P1A evidence review and saved-ledger checks.
- **E2.** v8 original-project reassessment report, collision results, exact theory certificate and scope review.
- **E3.** Local clean-training frozen protocol, report, results and bundled stabilized-source implementation.
- **E4.** v11 gate-mechanism report, metric summary and frozen protocol.
- **E5.** Native eight-stage experiment-series summary and final timescale report.
- **E6.** Interpolation-control report and subsequent finite-precision binding report.
- **E7.** Exploratory latest-value, range, moment, minimal and write/hold/query package documentation.